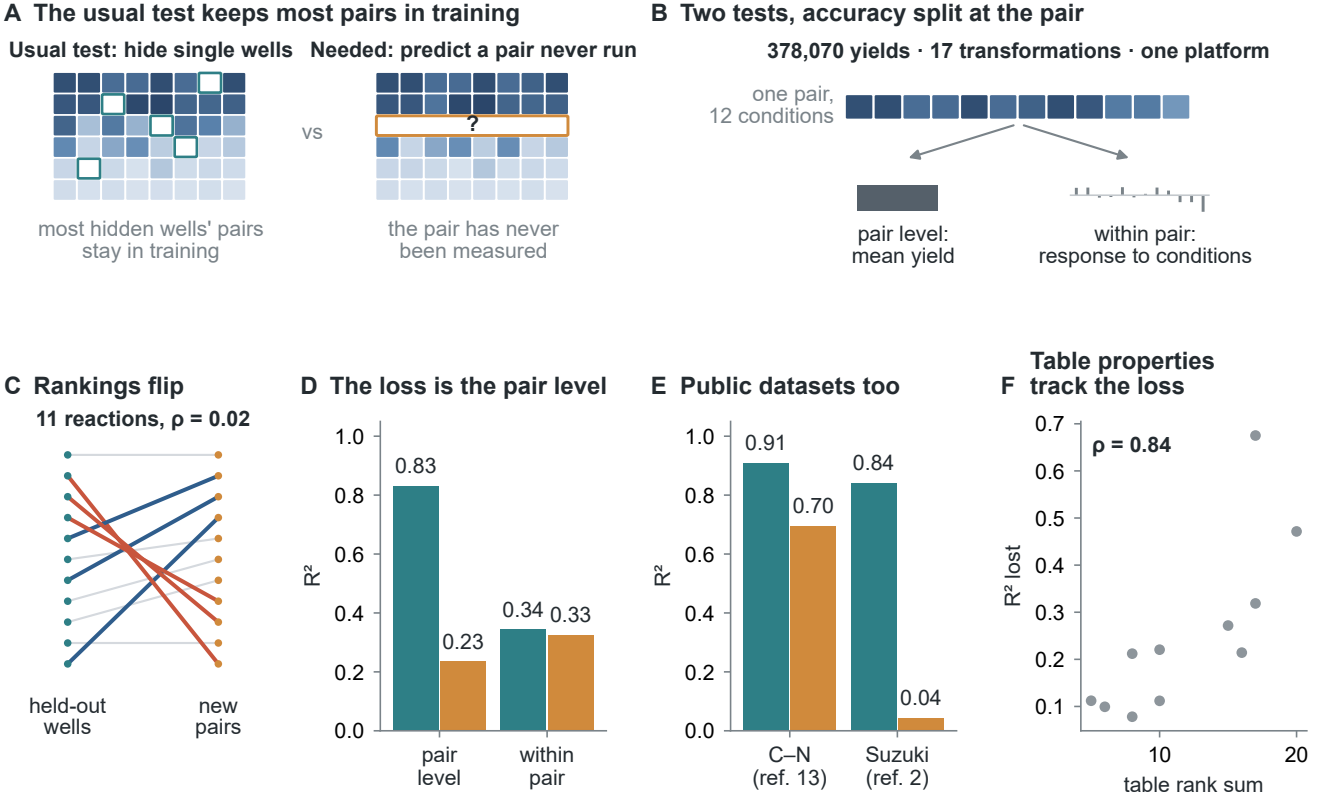

A The usual test keeps most pairs in training
Usual test: hide single wells
Needed: predict a pair never run
vs
?
most hidden wells' pairs stay in training
the pair has never been measured
B Two tests, accuracy split at the pair
378,070 yields · 17 transformations · one platform
one pair, 12 conditions
pair level: mean yield
within pair: response to conditions
C Rankings flip
11 reactions, ρ = 0.02
held-out wells
new pairs
D The loss is the pair level
R²
1.0
0.8
0.6
0.4
0.2
0.0
0.83
0.23
0.34
0.33
pair level
within pair
E Public datasets too
R²
0.91
0.70
0.84
0.04
C–N (ref. 13)
Suzuki (ref. 2)
F Table properties track the loss
ρ = 0.84
R² lost
0.7
0.6
0.5
0.4
0.3
0.2
0.1
10
20
table rank sum